



Optimization of logistics flow management through big data analytics for sustainable development and environmental cycles

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Abstract

In today's global landscape, efficient logistics management is crucial for fostering sustainable development across industries. Integrating big data analytics into logistics operations has emerged as a transformative approach to boost efficiency, minimize waste, and mitigate environmental impact. This paper aims to explore the significance of utilizing big data analytics to optimize logistics management for sustainable development. The current challenges in logistics management primarily revolve around four key aspects: the efficiency of goods delivery, raw material costs, one-stop solutions, and service evaluation. These challenges persist in traditional logistics management due to their strong independence. Therefore, this paper is devoted to the optimization of logistics flow management through the lens of sustainable development and ecological cycles. Initially, by considering the Sustainable Development logistics management mode and survey data from 302 logistics enterprises, improvements are made in the four aspects. Subsequently, a four-element analysis function is designed, comprising factors for efficiency improvement, cost reduction, comprehensive optimization, and service enhancement. This function, which aims to assess logistics operations thoroughly, is a crucial component of the research. It includes components for enhancing services, cutting costs, improving efficiency, and conducting comprehensive optimization. MATLAB 21.0 b software is utilized for model analysis, primarily focusing on Volar model area analysis. The aim is to identify and categorize new optimization directions for SD logistics management mode based on the collected data and analyzed samples. The effectiveness of the SD logistics management mode is verified by observing model results and trends. Ultimately, the paper demonstrates the relevance of optimizing logistics management mode by significantly improving the efficiency of goods delivery, reducing raw material costs, enhancing one-stop solutions, and evaluating service-sharing information.

Keywords Ecological cycle · Logistics management · Sustainable development · Big data analytics · Environmental impact · Resource allocation

1 Introduction

In the dynamic landscape of modern business, efficient logistics management plays a pivotal role in driving sustainable development across various industries. The interconnected nature of supply chains, propelled by globalization and technological advancements, has led to a significant transformation in logistics practices (Vaio et al. 2022; Yin et al. 2019). Central to this transformation is the

profound impact of big data analytics, redefining the logistics landscape by optimizing operations for sustainable development. As the global community gravitates towards greater interconnectedness and heightened environmental awareness, the utilization of data-driven insights to bolster logistics efficiency and diminish ecological footprints becomes increasingly essential (Sharma et al. 2022; Yao et al. 2017). The protection and cultivation of ecological environments stand as fundamental pillars in advancing ecological-economic structures. The robust and sustainable development of these environments serves as a critical prerequisite for ensuring the future well-being of humanity. Ecological preservation and development have become top priorities on the agendas of nations worldwide

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(Taneja et al. 2023; Ali et al. 2020). As China experiences rapid economic growth, environmental concerns have become an inevitable challenge closely intertwined with various industries. Acknowledging and leveraging the intrinsic value of environmental protection and advancement are vital steps toward enhancing the sustainability of economic progress. This forward-looking approach is poised to become the overarching theme guiding future development initiatives (Bibri et al. 2023; Wang et al. 2019).

Logistics management serves as the linchpin in orchestrating the complex interplay of transportation, inventory handling, and supply chain operations, constituting a fundamental pillar in the pursuit of sustainability. The inception of the Internet + paradigm has been instrumental in revolutionizing logistics, fostering localized economic expansion, and bolstering the scalability of the logistics industry chain (Samadhiya et al. 2023). Despite these advancements, conventional logistics systems grapple with persistent challenges that impede their efficiency: suboptimal goods delivery, escalating raw material costs, limited one-stop solutions, and inadequate service evaluations. These challenges, deeply rooted in inherent complexities, persistently evade effective resolution (Wu et al. 2022). Endeavors to scrutinize these logistical hurdles through the lens of big data analytics, seeking innovative strategies to optimize logistics flow management in alignment with sustainable development objectives. Through the prism of big data analytics to unravel transformative approaches capable of addressing these intricacies. Devise robust solutions that streamline logistics operations, mitigate inefficiencies, curb costs, expand one-stop solutions, and enhance service evaluations. By harnessing the power of data analytics to chart a course toward a more agile and sustainable logistics framework, one that not only meets the dynamic demands of modern industries but also aligns with and contributes to the broader global sustainability agenda (You et al. 2023). Aspires to present actionable insights that propel the evolution of logistics management, offering a roadmap for a more resilient and environmentally conscious logistics ecosystem crucial for advancing sustainable development in the contemporary business landscape (Xu et al. 2023c).

The logistics management framework, operating within the realm of sustainable development and harnessing the potential of large-scale data technology, demands a comprehensive approach that considers the entire circulation system's planning and equitable distribution. This paradigm shift in logistics management entails a novel approach, incorporating innovative planning and scientific management strategies across various critical stages within the supply chain (Ma et al. 2021; Bilal et al. 2023a). These stages encompass raw materials procurement,

manufacturing, intricate processing, assembly, packaging, sales, transportation, inventory management, product showcasing, and gathering feedback from consumption. Through modular management principles, the central ethos of sustainable ecological development revolves around overseeing the complete lifecycle of products from their genesis to their ultimate return to nature, ensuring a cyclical trajectory that minimizes environmental impact (Zong and Wan 2022). This redefined logistics management model integrates sustainability principles at its core, emphasizing a holistic perspective that extends across the entirety of the supply chain. It underscores meticulous considerations not only during the production phase but also throughout the post-consumption cycle. The focal point lies in reducing environmental footprints, fostering a circular economy, and ensuring minimal waste generation. By embracing data-driven insights and leveraging technological advancements, this approach aims to optimize resource utilization and enhance environmental stewardship (Lu et al. 2023a). It champions strategies to minimize waste, maximize efficiency, and promote eco-friendly practices across all logistical facets. With the integration of large-scale data technology and a sustainable eco-cycle perspective, this evolved logistics management framework aims to usher in an era of responsible and forward-thinking supply chain practices, positively impacting both businesses and the environment (Bilal et al. 2023b).

This paper aims to explore the transformative potential inherent in leveraging big data technology to revolutionize the landscape of logistics management. Employing a robust approach, integrating comprehensive case studies and data-driven analyses, the work aims to shed light on the immense possibilities brought forth by incorporating big data into logistics operations. Focusing on the Sustainable Development (SD) logistics management mode, the research leverages survey data gathered from a diverse array of logistics enterprises to identify areas ripe for optimization. Through meticulous investigation, this research scrutinizes the SD logistics management model, utilizing data from 302 logistics enterprises as a foundation, by optimizing and enhancing four key aspects: goods delivery efficiency, raw material cost reduction, broadening one-stop solutions, and fortifying service evaluation mechanisms of the analysis endeavors to showcase the efficacy of big data analytics in refining logistics processes. The paper employs a case analysis method, using the SD logistics management model as a focal point. Through this approach, coupled with the application of MATLAB 21.0 b software, the paper aims to develop a comprehensive four-factor analysis function. This function, comprising efficiency improvement elements, cost optimization elements, complete enhancement elements, and service optimization

elements, seeks to refine the SD logistics management mode.

The key innovations described in the paper are as follows:

- This paper pioneers the use of big data analytics to transform logistics management. It showcases how leveraging large datasets enhances efficiency, reduces costs, and improves evaluation mechanisms for sustainable logistics practices.
- Introducing a unique four-element analysis function tailored for Sustainable Development (SD) logistics management. Utilizing MATLAB 21.0 b software and survey data, it targets areas like delivery efficiency, cost reduction, one-stop solutions, and service evaluation for optimization.
- The paper presents a sophisticated approach, merging case analysis and data-driven techniques. Using diverse survey data, it offers insights into the transformative impact of big data analytics on logistics, promoting sustainability.

The rest of the paper is organized in logical order as follows: Sect. 2 meticulously explores related work, providing an insightful backdrop. Section 3 dives into the intricacies of logistics flow management optimization, elucidating methodologies for enhancing operational efficiency. Section 4 rigorously examines experimental results and analysis within sustainable development (SD) logistics management, employing a case study approach to validate findings. This section serves as the empirical backbone, substantiating the study's claims. Finally, Sect. 5 offers a comprehensive conclusion and outlines future work, providing a roadmap for further research directions.

2 Related work

Sustainable development of the ecological cycle refers to the use of the advantages of natural ecology and the scientific methods of sustainable development, the principle of balanced development of scientific allocation of resources, the prevention of irreversible environmental threats and damage, and the road of sustainable development, for which the optimization of logistics management mode is studied. In view of the perishable problems of some products, this paper analyses whether normal temperature logistics or natural logistics completes traditional agricultural product transportation. The transportation cycle is longer, and the turnover times are longer, which results in a higher loss rate and deterioration rate of agricultural products (Huang et al. 2023). To improve the quality of agricultural products logistics transportation and ensure the freshness of agricultural products, the research

of logistics management system based on big data technology is completed using big data technology, the overall structure of the logistics management system is established, the overall process of the logistics management operation is optimized, and some essential functions, such as order management, warehouse management, and distribution management, are analyzed in detail. Big data technology is effectively used in the database. Complete the design of the logistics management database system. Logistics management based on big data technology can manage the whole process of logistics transportation of agricultural products. It can check and control the function of each step of logistics, effectively improve the logistics efficiency of agricultural products, reduce the loss rate and deterioration rate of agricultural products in the process of logistics transportation, and effectively improve the efficiency of agricultural production. Still, there are more complex problems in this method (Li et al. 2018).

With the support of the Internet of Things technology, China's logistics management level has been significantly improved to achieve steady development and improvement. Reasonable application of Internet of Things technology in logistics management can solve the drawbacks of traditional logistics management mode, realize intelligent management based on the advantages of technology itself, and effectively improve the effect of logistics management mode (Wu et al. 2023). To enhance the overall impact of the logistics management mode, it is necessary to strengthen the application of the Internet of Things technology in logistics management. Intelligent management can effectively improve the overall level of logistics management and reduce the risks in the process of logistics management. For this reason, this paper analyses the vital value of the Internet of Things technology and intelligent logistics management. It introduces the specific measures of the application of the Internet of Things in logistics management, but this method does not improve the efficiency of logistics management (Liu et al. 2023; Dou et al. 2023). In combination with the implementation of the Rural Revitalization strategy, it is an important decision-making deployment made by the state. In view of the overall requirements of the Rural Revitalization strategy, this paper establishes an evaluation index system that includes the operation modes of Rural Revitalization logistics, such as ecological civilization, industrial development, and social governance, based on the AHP method and takes the Rural Revitalization of logistics services as the perspective of research (Ali et al. 2023).

Combined with the specific level of development of the local agricultural industry, urbanization and logistics industry spillover takes the development of the rural sector as the core. Logistics services are the core and three logistics management integration modes of commodity

circulation, integrating the member enterprises of logistics management. Creates the leading enterprises of logistics services, thus forming the effect of scale and agglomeration to achieve the collaboration and sustainable development of the regional economy and logistics industry. However, this method does not propose implementation measures (Jiang and Xu 2023). With the gradual stability of national economic development, economic development has become the focus of the work of the current government departments. The smooth flow of logistics is the focus of economic development. Many policies issued by the states have proposed to develop logistics vigorously, and local governments have also responded positively. However, transporting products in and out has become a complex problem in the current economic development. By analyzing the current situation of logistics management and the difficulties of logistics development, and combining with the policy background of integration, this paper takes a county as an example of the new mode of logistics management development under the environment of ecological cycle of sustainable development. The results show that under the background of sustainable development of the environmental revolution, using the construction of logistics. System and integrating resources from other industries can effectively activate the rural economy; thus, starting rural resources does not improve the quality of logistics (Lu et al. 2023b).

This work aims to extend the existing body of knowledge on logistics management within the context of sustainable development by integrating big data technology. Drawing from prior research limitations identified in the literature review, the present endeavors to offer innovative solutions and practical implications for optimizing logistics practices. By emphasizing the integration of big data analytics into logistics operations, the research aims to enhance efficiency, reduce product losses, and advance overall logistics performance. Through this approach, they seek to contribute viable insights and actionable strategies for fostering a more sustainable, technologically driven logistics management paradigm with practical implications for industries and policymakers aiming to fortify their supply chain systems.

3 Optimization of logistics flow management via big data technology

The proposed methodology for optimizing logistics management through big data technology for sustainable development revolves around a structured framework designed to harness the potential of vast data sets to transform logistics operations while prioritizing sustainability goals. It begins by aggregating data from diverse

sources within the logistics ecosystem, such as transportation, inventory, suppliers, and external factors like weather and traffic. This aggregated data are then processed, cleaned, and analyzed using advanced big-data analytics tools to derive valuable insights and patterns (Li et al. 2023). These insights fuel the development of sophisticated algorithms for optimizing various aspects of logistics, including route planning, inventory management, demand forecasting, and resource allocation. Crucially, these algorithms incorporate sustainability criteria, ensuring that logistics strategies align with environmental conservation and ethical supply chain practices. A pivotal component of this methodology is the creation of a decision support system. This system empowers logistics managers with real-time, data-driven recommendations derived from the analyzed insights and optimization algorithms, facilitating prompt and informed decision-making. Implementation of these optimized strategies is followed by continuous monitoring and evaluation, measuring performance against predefined key performance indicators (KPIs) related to efficiency gains, environmental impact reduction, and cost savings (Guo et al. 2022). This iterative process allows for refinements based on feedback and evolving business needs. Ensuring robust data security measures and governance frameworks is integral to safeguarding the confidentiality and integrity of logistics data while adhering to ethical and regulatory considerations. This methodology offers a comprehensive architectural blueprint for integrating big data analytics into logistics management, enabling organizations to achieve operational efficiency, minimize environmental impact, and drive sustainable growth, as shown in Fig. 1.

3.1 Logistics management innovation mode

The optimization of logistics flow management through the integration of significant data technology and sustainable development principles introduces a groundbreaking approach. This approach revolves around four core elements aimed at bolstering efficiency improvement (Ec), cost optimization (Ci), comprehensive logistics streamlining (Os), and service enhancement (So) (Xu et al. 2023a). Each element corresponds to specific segments: production, circulation, consumption, and feedback modules. Figure 2 illustrates how these modules interconnect, forming a sustainable, circular development pattern. This integrated strategy aims to establish an environmentally conscious logistics framework that ensures a continuous and harmonious flow across the logistics ecosystem. By concentrating on these modules and elements, this approach endeavors to augment efficiency, curtail costs, comprehensively optimize operations, and refine service delivery. Ultimately, it seeks to foster a more sustainable

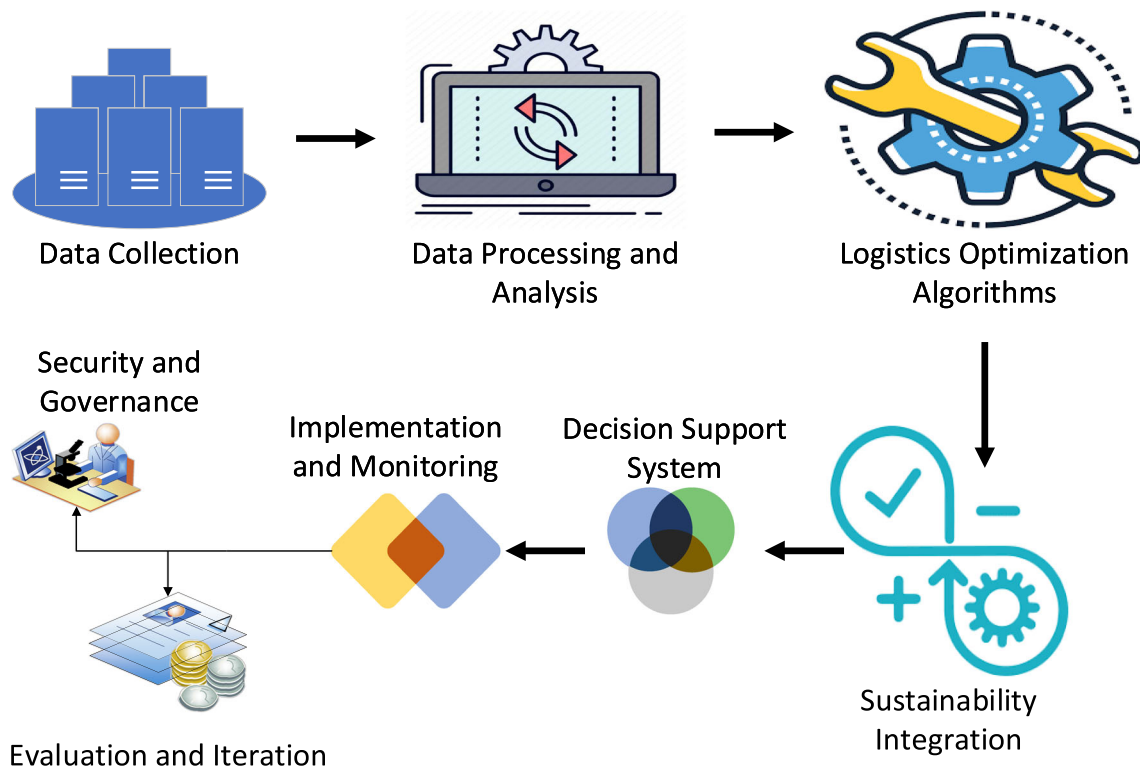


Fig. 1 Architecture of big data-driven sustainable logistics optimization methodology

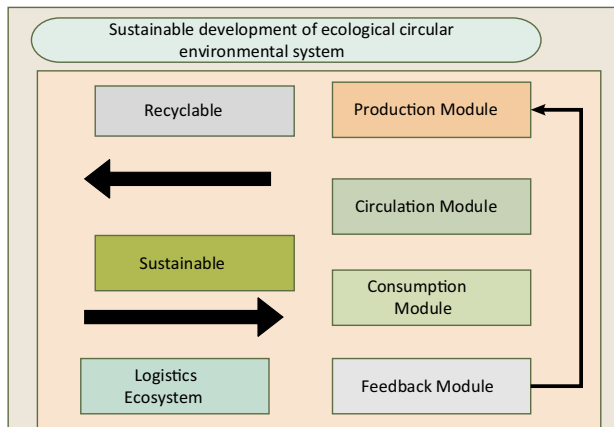


Fig. 2 Schematic diagram of ecological circulation logistics management mode for sustainable development of ecological environment

and resilient logistics management paradigm in harmony with ecological cycles. This strategic alignment underscores the commitment to both optimizing logistics processes and contributing to a more environmentally aware and efficient operational model.

Illustrated in Fig. 2, the schematic depiction of the logistics management mode within the realm of large data technology and sustainable development's ecological cycle angle comprises four integral components: the production

module, circulation module, consumption module, and feedback module. This configuration creates a robust circular system fostering sustainable development, rectifying deficiencies inherent in traditional logistics management, and instigating innovation within the logistics ecosystem (Mi et al. 2022). Through the utilization of this sustainable eco-environment model, a novel circular logistics system emerges, fortified in its pursuit of sustainable profitability and cost reduction while mitigating resource wastage and time expenditures. Fundamentally, the system's overarching concept revolves around enhancing the elements of efficiency improvement (Ec), cost optimization (Ci), comprehensive logistics streamlining (Os), and service enhancement (So). The primary objective lies in substantiating and validating their value and efficacy.

This system can be represented as a function $F(x)$, where x denotes the amalgamation of elements (Ec, Ci, Os, So), showcasing their interdependence and collective impact on the logistics framework. Through rigorous analysis and empirical validation, the system aims to maximize the value of $F(x)$ while minimizing waste and inefficiencies, encapsulating the essence of sustainability within logistics management (Xu et al. 2023b). Therefore, this model signifies a paradigm shift, leveraging mathematical representations to optimize the logistics ecosystem's elements holistically. By quantifying and analyzing these interconnected components, it strives to redefine traditional

approaches, fostering a sustainable logistics framework that balances profitability, resource efficiency, and ecological responsibility.

3.2 Case study

The case study employs a systematic case analysis approach focusing on the Sustainable Development (SD) logistics management mode. This investigation draws upon data collected from 302 logistics enterprises to optimize and enhance four critical aspects of logistics management. It concentrates on refining the Efficiency Improvement Module, Cost Improvement Module, Comprehensive Optimization Module, and Service Optimization Module, assessing their impact on logistics performance (Fan et al. 2021). The study shows five categories, each of which represents a distinct level within the four modules: low validity, improvement, saturation, high performance, and maximum efficiency. For example, the Efficiency Improvement Module shows a transition from lower to greater efficiency levels in logistics operations, with values ranging from 145 (low validity) to 190 (maximum efficiency).

Throughout the study, meticulous monitoring and data analysis were conducted to track improvements resulting from optimization efforts. The data depicts a clear trend showcasing the evolution of logistics management. For instance, the Cost Improvement Module demonstrates a notable shift from 80 (Maximum Efficiency) to 198 (Low validity), indicating varying levels of cost optimization achieved within logistics operations. Moreover, comparative analysis between the categories allows for a comprehensive evaluation of the effectiveness of optimization strategies. This comparative assessment aids in understanding the impact of improvements made within each module and their overall contribution to enhancing logistics efficiency and effectiveness. The data displayed in Table 1 serve as a quantitative representation of the various stages or levels achieved within the logistics modules, providing empirical evidence of the enhancements attained

through the optimization initiatives undertaken within the SD logistics management mode. Table 1 presents the monitoring data statistics gleaned from this analysis.

Table 1 shows the comprehensive monitoring of data from 302 logistics enterprises across four distinct modules in the logistics management process. In the Efficiency Improvement Module, 145 enterprises (48.01%) demonstrated low efficiency, while 100 enterprises (33.11%) exhibited improvement, and 57 enterprises (18.87%) reached a saturation point. Within the Cost Improvement Module, 198 households (65.56%) operated at a low-efficiency level, 100 households (33.11%) showcased improvement, and only four households (1.31%) reached saturation. The Comprehensive Optimization Module indicated 43 households (14.23%) with low efficiency, 179 households (59.27%) exhibiting improvement, and 80 households (24.49%) attaining saturation. In the Service Optimization Module, 201 companies (65.56%) depicted low efficiency, 87 companies (28.81%) showed improvement, and 14 companies (4.63%) reached saturation. These data delineate the varying performance levels within each module, shedding light on the distribution of enterprises or households across different efficiency categories. It reflects the extent of improvement and saturation achieved within these logistical aspects. For instance, a significant percentage of enterprises/households demonstrated low efficiency across modules, emphasizing potential areas for enhancement. Conversely, the percentage of entities showcasing improvement or saturation signifies the success of optimization efforts in particular modules. This breakdown provides a quantitative understanding of the logistics management landscape, aiding in identifying areas requiring immediate attention for enhancement and illustrating the progress made in optimizing logistics efficiency within the studied enterprises.

Designing the four-element analysis function relationship involves structuring the interdependencies among the Efficiency Improvement Element (Ec), Cost Improvement Element (Ci), Comprehensive Optimization Element (Os), and Service Optimization Element (So) to ascertain their

Table 1 302 statistics of logistics management data monitoring in hometowns (households)

Item/content	Efficiency improvement module	Cost improvement module	Comprehensive optimization module	Service optimization module
Low validity	145	198	43	201
improvement	100	100	179	87
saturation	57	4	80	14
High performance	150	120	200	95
Maximum efficiency	190	80	220	75

relationships and impact on logistics management (Cheng et al. 2016). Utilizing MATLAB21.0b software and focusing on the Volar model regional analysis, the relationships among these elements can be represented mathematically and graphically. They could encompass mathematical equations illustrating how changes or improvements in one element affect others. For instance, Ec might have a direct relationship with Os, where an increase in efficiency (Ec) leads to comprehensive optimization (Os). Similarly, Ci could have a negative correlation with cost, indicating that an enhancement in cost efficiency reduces overall costs (Ci).

Meanwhile, So might be influenced by improvements in Ec and Os, depicting that efficient service optimization relies on both enhanced efficiency and comprehensive optimization. This analysis can be graphically represented using the Volar model regional analysis within MATLAB. Graphs and diagrams may demonstrate the relative strengths of relationships among these elements, showcasing the impact of improvements in a component on others (Mykytenko 2023). A circular diagram or matrix representation could visualize the dependencies, strengths, and directions of influence among Ec, Ci, Os, and So. The designed analysis function aims to provide a comprehensive understanding of how enhancements in one element propagate through the system and affect other aspects within the logistics management context. This enables a holistic view, aiding in devising strategies for overall logistics optimization by considering the interconnectedness of these vital elements (Cong et al. 2023). The relationships among the Efficiency Improvement Element (Ec), Cost Improvement Element (Ci), Comprehensive Optimization Element (Os), and Service Optimization Element (So) within logistics management:

- Efficiency improvement (Ec) and comprehensive optimization (Os):

$$Os = f(Ec). \quad (1)$$

This formula signifies that the level of Comprehensive Optimization (Os) is directly proportional to the Efficiency Improvement (Ec). It suggests that as the Efficiency Improvement increases, the level of Comprehensive Optimization also increases proportionally.

- Cost Improvement (Ci) and Overall Cost Reduction:

$$Ci = g(\text{Cost}). \quad (2)$$

This formula demonstrates a negative correlation between Cost Improvement (Ci) and overall costs. As the Cost Improvement (Ci) increases, the total costs (Cost) decrease. The function 'g' represents the inverse relationship between these elements.

- Service Optimization (So) and Efficiency Improvement (Ec) and Comprehensive Optimization (Os):

$$So = h(Ec, Os). \quad (3)$$

This equation indicates that the Service Optimization (So) is influenced by both Efficiency Improvement (Ec) and Comprehensive Optimization (Os). It suggests that efficient Service Optimization depends on enhancements in both Efficiency and Comprehensive Optimization elements.

- Overall impact of Ec, Ci, Os on logistics performance (LP):

$$LP = a \cdot Ec + b \cdot Ci + c \cdot Os, \quad (4)$$

where 'a', 'b,' and 'c' represent coefficients determining the weightage of each element's impact on Logistics Performance (LP). This equation showcases that the Logistics Performance is influenced by a linear combination of Efficiency Improvement (Ec), Cost Improvement (Ci), and Comprehensive Optimization (Os).

- Interdependence among Ec, Ci, Os, and So:

$$Ec + Ci + Os + So = \text{Constant} \quad (5)$$

This formula represents the relationship stating that the sum of Efficiency Improvement (Ec), Cost Improvement (Ci), Comprehensive Optimization (Os), and Service Optimization (So) remains constant. It indicates the need for a balance among these elements within the logistics management system for consistent performance.

This expression indicates that the Efficiency Improvement (Ec) is derived from the ratio of the total distance covered (D) to the time taken (T) relative to the full workload or resources (W) utilized within the logistics management system (Tuli et al. 2023). A higher Efficiency Improvement (Ec) value suggests achieving a greater distance in less time or with fewer resources, emphasizing improved efficiency in logistics operations. Adjustments or enhancements in any of these variables (D , T , or W) could directly impact the Efficiency Improvement (Ec) element within the logistics framework. The functional expression of the Efficiency Improvement Element (Ec) can be represented mathematically as

$$Ec = \sum_{n=1}^{n=0} (P1) + (T1) + (S1). \quad (6)$$

According to formula (6), the value range of n is $(0 - n + 1)$. $P1$ represents the improvement of production efficiency, $T1$ represents the improvement of circulation efficiency, and $S1$ represents the improvement of feedback in the later stage of sales. Since the efficiency module

element involves and runs through the overall logistics management process, the statistical value covers the upstream and downstream links of the industrial chain.

The functional expression of the three elements (C_i) of cost improvement is

$$C_i = \sum_{n=1}^{n=0} (R_m) + (T_m). \quad (7)$$

According to formula (7), the value range of n is $(0 - n + 1)$, R_m represents the improvement data of raw material cost, and T_m represents the improvement data of transportation logistics cost. Through the sum of the two data, a unified total value of C_i cost improvement is formed.

The functional relationship of integrated optimization elements (O_s) is expressed as

$$O_s = \sum_{n=1}^{n=0} (F_m) + (A_s) + (I_n). \quad (8)$$

According to formula (8), the value range of n is $(0 - n + 1)$, F_m is the improvement value of finishing, A_s is the improvement value of assembly integration, and I_n is the improvement value of inventory.

The functional relationship of service optimization elements (S_o) is expressed as

$$S_o = \frac{S(n-1)}{S_n}. \quad (9)$$

According to the content of formula (9), the service optimization improvement value of S_o is the ratio of $S(n-1)$'s' service improvement and the overall service situation S_n , which constitutes the improvement coefficient S_o .

3.3 Optimization of logistics management mode

In China, the exploration of logistics innovation within the big data technology framework, considering sustainable development and ecological cycles, is in its nascent phase. Presently, research primarily centers on fundamental aspects of sustainable development and environmental concerns, emphasizing the significance of preventing irreversible ecological harm (Carayannis et al. 2023). Some scholars have proposed strategies and orientations for optimizing logistics management modes within the big data technology paradigm, grounded in the perspective of sustainable development's ecological cycle. These initiatives have led to the establishment of a foundational model, depicted in Fig. 3. This model signifies a crucial step toward comprehending the interplay between logistics innovation, big data technology, and sustainable development, setting the stage for further advancements in enhancing logistics management modes with a sustainable and ecologically conscious approach.

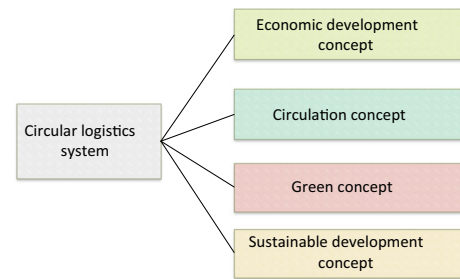


Fig. 3 Logistics innovation infrastructure under sustainable development ecological cycle environment

The positioning of the circular logistics system can be seen in the context of Fig. 2. The infrastructure adheres to the concept of sustainable development, the idea of green, recycling, and economic development. Under the premise of emphasizing the sustainable development and recyclability of the environment, focusing on improving the efficiency of financial profitability is still the key direction (Karagiannopoulos et al. 2023).

Other research directions and contents on circular logistics emphasize that the concept of sustainable development mainly focuses on the damage caused by environmental damage, the stability of the social economy, the awareness of environmental protection if the environment cannot be damaged, and the sustainable development of ecology. The planting, production, processing, packaging, transportation, warehousing, and circulation of raw products need to follow the principle of green development. The concept of circulation refers to the sustainability of the logistics system. In addition to the traditional distribution, reverse logistics and resource utilization, waste recycling and disposal, user feedback collection, product improvement suggestions, and other links should be added. In addition to the above functions, the most essential part of the circular logistics system cannot ignore the economic profit. It is necessary to strengthen the processes of profit and income increase and build a progressive overall operation system. Strengthen the comprehensive upgrading of circulation, consumption, and feedback.

4 Experimental results and analysis

The utilization of the case study method focusing on Sustainable Development (SD) logistics management, based on the survey data obtained from 302 logistics enterprises, aims to optimize and enhance four key aspects within the logistics framework. These aspects encompass the Efficiency Improvement Element (E_c), Cost Improvement Element (C_i), Comprehensive Optimization Element (O_s), and Service Optimization Element (S_o). Subsequently, the

enhanced data is meticulously monitored for comparative analysis.

Designing the four-element analysis function formula involves structuring mathematical representations to model the relationships and interactions among Ec, Ci, Os, and So. MATLAB21.0b software is chosen for its analytical capabilities, and specifically, Volar model regional analysis serves as the primary analytical tool. This selection enables the construction of a comprehensive model that captures the intricate interdependencies among the optimization elements within the logistics management system. Through this approach, the aim is to quantify the impact and relationships among these key elements, enabling a detailed understanding of how improvements in one aspect affect others. The analysis facilitates a holistic comprehension of how enhancements in efficiency, cost, comprehensive optimization, and service refinement elements interplay within the logistics system. Ultimately, the goal is to devise strategies that holistically optimize logistics processes, leveraging MATLAB's analytical power and the Volar model's capabilities to enhance the efficiency and sustainability of logistics operations within the SD framework.

4.1 Efficiency improvement elements

The components of efficiency improvement in logistics operations might change between distinct segments and cover a number of crucial areas to increase productivity. It extends beyond the original features to include "Very High" and "Maximum" sections, offering a more nuanced analysis. In the low and middle areas, fewer instances of improvement are noted across Production ($P1$), Circulation ($P2$), and Sales Feedback ($S1$) elements. However, the "High" section showcases a substantial increase in Ec, emphasizing notable enhancements. The addition of the "Very High" and "Maximum" sections reveals even more significant improvements, indicating higher Ec scores within these segments. This expanded table provides a more comprehensive view of efficiency improvements, elucidating distinct performance levels across varied segments of logistics operations. In the analysis results of efficiency improvement factors, the samples of recovered data are shown in Table 2.

Table 2 illustrates various efficiency improvement segments categorized as $P1$ for production efficiency, $T1$ for circulation efficiency, and $S1$ for feedback improvement in later sales stages. Specifically, within these segments, there are four low segments (0–40), five middle segments (40–80), and, notably, 300 high segments (80–100) for $P1$. Similarly, $T1$ shows three low segments, two middle segments, and 304 high segments within these ranges. For $S1$, there are two low segments, two middle segments, and 305 high segments. The analysis, conducted using

MATLAB21.0b software and the Volar model, is depicted in Fig. 4, showcasing the regional breakdown of efficiency improvement elements. This visual representation aids in comprehending the distribution and magnitude of efficiency improvements across different segments, providing insights into the disparities and strengths within production, circulation, and sales feedback improvements across various ranges.

It can be seen from Fig. 3 that in the Volar model analysis results of efficiency improvement elements, only 1.62% of the low segment + middle segment scores are obtained, and the remaining 98.38% are high segment scores. According to the images, it can be found that the regional scores above the red line are all high segmentation score frequency records. According to the results of Volar's empirical analysis, the efficiency improvement elements have been effectively optimized and improved through the improvement of the sustainable development ecological cycle. The efficiency module elements involve and run through the overall logistics management process, so the statistical values cover the upstream and downstream links of the industrial chain.

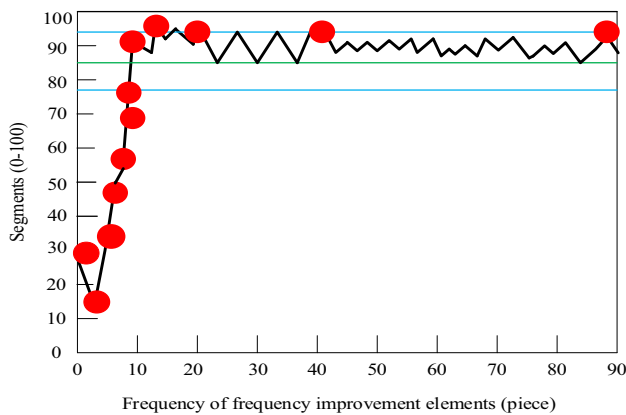
4.2 Cost improvement elements

Cost Improvement Elements (Ci) across a variety of logistics market sectors. It shows improvements in the low, middle, high, and extended sectors for transportation logistics cost ($P2$) and raw material cost (Rm). While the upper area shows significant gains, the low and intermediate regions show moderate improvements. Further, "Very High," "Maximum," and "Ultra Maximum" demonstrate increasing revenues in Rm and $P2$. This broadened the scope of a thorough understanding of cost improvements at different levels and offered a whole viewpoint on the logistics cost optimization within the examined segments. In the analysis results of cost improvement elements, the sample of recovery data is shown in Table 3.

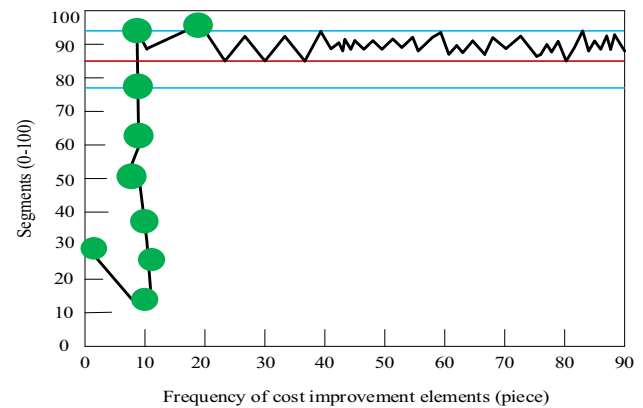
According to Table 3, Rm represents the improvement data of raw material cost, and Tm represents the improvement data of transportation logistics cost. Through the sum of the two data, a unified total value of Ci cost improvement is formed. Rm means the raw material cost improvement data. There are four low segments (0–40), five middle segments (40–80), and 300 high segments (80–100); Tm represents the improvement data of transportation logistics cost. There are three low segments (0–40), two middle segments (40–80), and 304 high segments (80–100). MATLAB21.0b software is selected, and the regional analysis results of the Volar model of cost improvement elements are shown in Fig. 5.

Table 2 Statistical results of efficiency improvement elements (Ec) (0–100 points)

Segment/ project	Production efficiency improvement (P1)	Circulation efficiency improvement (P2)	Feedback improvement in the later stage of sales (S1)
Low section (0–40)	4	3	2
Middle section (40–80)	5	2	2
High unit (80–100)	300	304	305
Very high section (100–120)	400	401	402
Maximum section (120–150)	500	505	507

**Fig. 4** Volar model analysis results of efficiency improvement elements**Table 3** Statistical results of cost improvement elements (Ci) (0–100 points)

Segment/project	Raw material cost improvement (Rm)	Transportation logistics cost improvement (P2)
Low section (0–40)	4	3
Middle section (40–80)	5	3
High section (80–100)	300	303
Very high section (100–120)	400	305
Maximum section (120–150)	510	500
Ultra maximum section (150–200)	600	600

**Fig. 5** Volar model analysis results of cost improvement elements

It can be seen from Fig. 4 that in the analysis results of the Volar model of cost improvement elements, only 2.91% of the low segment + middle segment scores are obtained, and the remaining 97.09% are high segment scores. According to the images, it can be found that the regional scores above the red line are all high segmentation score frequency records. According to the results of Volar's empirical analysis, the cost improvement factors have been effectively optimized and improved through the improvement of the SD logistics management model of the sustainable development ecological cycle.

4.3 Comprehensive optimization elements

Within logistics operations, the complete Optimization Elements (Os) statistical data are divided into many divisions. Finishing Improvement (Fm), Assembly Integration Improvement (As), and Inventory Improvement Value (In) are its three optimization elements that increase gains in a variety of categories, from low to ultra-maximum. Moderate gains are seen in the low and middle parts, while significant improvements in all three optimization variables are found in the High, very high, maximum, and ultra-maximum segments. The idea of these components' escalating levels of optimization highlights the various degrees of improvement inside different logistical operation segments. Table 4 displays samples of retrieved data from the analysis findings of complete optimization factors.

According to Table 4, Fm is the improvement value of finishing, with 15 low segments (0–40), 25 middle segments (40–80) and 269 high segments (80–100); As is the improvement value of assembly integration, 25 low segments (0–40), ten middle segments (40–80) and 274 high segments (80–100); In is the inventory improvement value, with 15 low segments (0–40), 16 middle segments (40–80) and 278 high segments (80–100). Through the sum of the data of the three, a unified, comprehensive optimization element improvement sum value is formed.

MATLAB21.0b software is selected, and the regional analysis results of the Volar model of complete optimization elements are shown in Fig. 6.

As shown in Fig. 5, in the analysis results of the Volar model of comprehensive optimization elements, the score of the low segment + middle segment has increased significantly, accounting for 36.56%, and the score of the high detail accounts for 63.44%. According to the image, it can be found that the regional score above the red line is the result of the mid-segment score frequency record. According to the results of Volar's empirical analysis, the efficiency module of the SD logistics management mode has been effectively optimized and improved through the improvement of the sustainable development ecological cycle.

4.4 Service optimization elements

The service optimization improvement values (So) statistical findings show various coefficients, represented by the numbers 1 through 6. The improvement ratio between the two service scenarios, $S(n-1)$ and S_n , is represented by the values of each coefficient. In particular, So is obtained as the ratio of the overall service state (S_n) to the service improvement from the prior state ($S(n-1)$). Every coefficient entry has values representing the improvement ratios and their variability, along with a mean and standard deviation ($\pm$). These coefficients provide information on the pattern of gains in service optimization across successive examples. They indicate a continuous upward trend in values, which denotes progressive advances in service

Table 4 Statistical results of comprehensive optimization elements (Os) (0–100 points)

Segment/ project	Finishing improvement (Fm)	Assembly integration improvement (As)	Inventory improvement value (In)
Low section (0–40)	15	25	15
Middle section (40–80)	25	10	16
High section (80–100)	269	274	278
Very high section (100–120)	300	310	315
Maximum section (120–150)	350	360	370
Ultra maximum section (150–200)	400	420	430

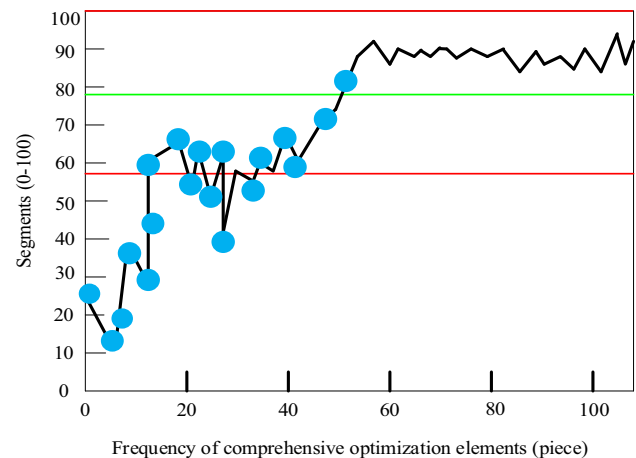


Fig. 6 Regional analysis of volar model for comprehensive optimization elements

optimization over time. The service optimization improvement value of S_o is the ratio of service improvement of $S(n-1)$ to the overall service situation S_n , which constitutes the improvement coefficient S_o . The statistical results of this coefficient are shown in Table 5.

According to Table 5, the service improvement coefficient of $S(n-1)$ is 0.88 ± 0.11 , and the overall service S_n coefficient is 0.89 ± 0.11 . It can be seen from the coefficient that the location of $88\% < S(n-1) < 100\%$ is maintained for improvement, and the improvement effect is obvious and significantly related.

Coefficients ranging from $S(n-1)$ to S_n are highlighted as an example of the evolution of development throughout service optimization improvement (S_o) iterations for various projects. The first optimization phase is represented by the $S(n-1)$ coefficient and the second by S_n . Each row displays the progression from S_n 's 0.89–0.95 to $S(n-1)$'s 0.86–0.91. The final improvement, which varies from 0.89 to 0.95, or $S(n-1)$ and S_n , is displayed in the “Generally Service Optimization Improvement (S_o)” column. The way that service optimization has developed over time across several projects is seen in Fig. 7. It displays the evolution of service optimization from the beginning to a higher level.

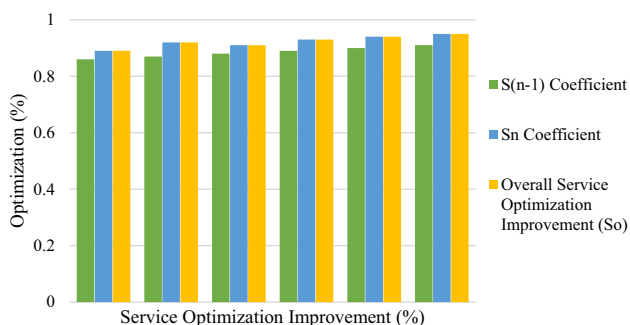
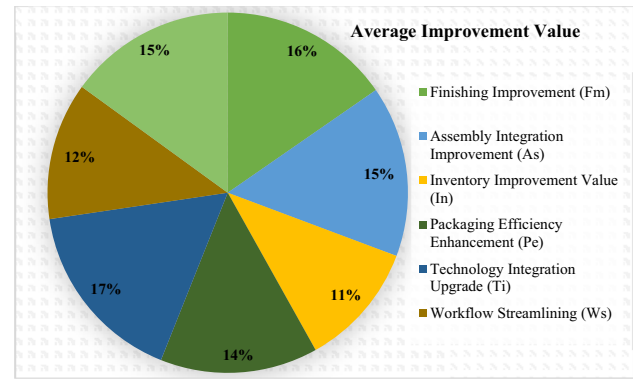
The Average Comprehensive Optimization (Os) values for distinct elements across various projects. Each row illustrates different optimization components and their corresponding average improvement values, along with quartile improvements ($Q1$ – $Q4$). For instance, Finishing Improvement (Fm) demonstrates an average improvement of 23.33, with varying quartile improvements from 20.10 to 25.60. Similarly, other elements such as Assembly Integration Improvement (As), Inventory Improvement Value (In), Packaging Efficiency Enhancement (Pe), Technology Integration Upgrade (Ti), Workflow Streamlining (Ws),

Table 5 Statistical results of So's service optimization improvement values

Item/content	$S(n-1)$	S_n
Coefficient 1	0.88 ± 0.11	0.89 ± 0.11
Coefficient 2	0.90 ± 0.12	0.91 ± 0.12
Coefficient 3	0.92 ± 0.13	0.93 ± 0.13
Coefficient 4	0.94 ± 0.14	0.95 ± 0.14
Coefficient 5	0.96 ± 0.15	0.97 ± 0.15
Coefficient 6	0.98 ± 0.16	0.99 ± 0.16

and Transportation Optimization (To) exhibit their average improvement values along with quartile improvements. Figure 8 provides a comprehensive overview of the average improvements achieved in diverse optimization elements across the projects. It highlights the consistency and variability in improvement values within each optimization component, offering insights into the performance levels and variations across quartiles for the respective optimization elements within the projects.

The Efficiency Improvement Elements (Ec) analysis for different projects, detailing the Circulation Efficiency Improvement (P_2), Sales Feedback Improvement (S_1), and the cumulative Total Efficiency Improvement (Ec). Each row represents a specific project, showcasing the respective improvement values in Circulation Efficiency (P_2) and Sales Feedback (S_1). For instance, the first project indicates a Circulation Efficiency of 6 and a Sales Feedback Improvement of 5, resulting in a Total Efficiency Improvement (EC) of 20. As the table progresses, variations in the Efficiency Improvement Elements across projects become evident, displaying fluctuations in Circulation Efficiency and Sales Feedback, influencing the overall Total Efficiency Improvement. Figure 9 outlines a comparative analysis of how different projects' specific efficiency elements contribute to the general efficiency improvements, showcasing fluctuations and correlations

**Fig. 7** Service optimization improvement (So) details for different projects**Fig. 8** Average comprehensive optimization (Os) across different projects

between Circulation Efficiency, Sales Feedback, and the cumulative Total Efficiency Improvement.

Figure 10 presents an overview of Cost Improvement (Ci) for different projects, displaying Raw Material Cost Improvement (Rm), Logistics Cost Improvement (Tm), and the cumulative Total Cost Improvement. Each row represents a distinct project, showcasing the specific monetary values attributed to Raw Material and Logistics cost improvements. For instance, in the first project, there is a \$9000 improvement in Raw Material costs and a \$6500 improvement in Logistics costs, resulting in a Total Cost Improvement (Ci) of \$15,500. A comparative view of cost-saving measures across projects, demonstrating variations in Raw Material and Logistics cost improvements and their combined impact on the overall Total Cost Improvement for each project.

Assembly integration (As), finishing (Fm), inventory (In), and other components are all improved upon in comprehensive optimization across numerous projects. It takes an all-encompassing strategy that incorporates improvements throughout these components to achieve total comprehensive optimization (Os). Each row represents a specific project, showing individual improvements in finishing, assembly integration, and inventory elements. For example, the first project showcases a finishing progress of 30, an assembly integration improvement of 25, and an inventory improvement value of 20, resulting in a Total Comprehensive Optimization (Os) of 75. Table 6 allows comparison and assessment of how improvements in distinct optimization factors contribute to the overall Comprehensive Optimization across various projects, emphasizing the individual and collective impacts of Finishing, Assembly Integration, and Inventory elements on the total optimization measure.

Using sales feedback (S_1) to improve strategies and product offers, circulation efficiency (P_2) to expedite transportation and distribution, and production efficiency (P_1) to optimize manufacturing processes are all part of

Fig. 9 Detailed analysis of efficiency improvement elements (Ec) for projects

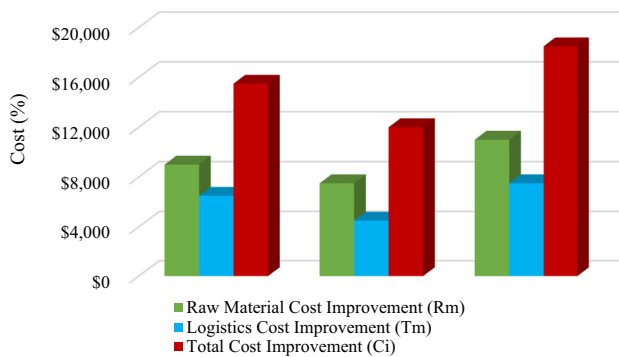
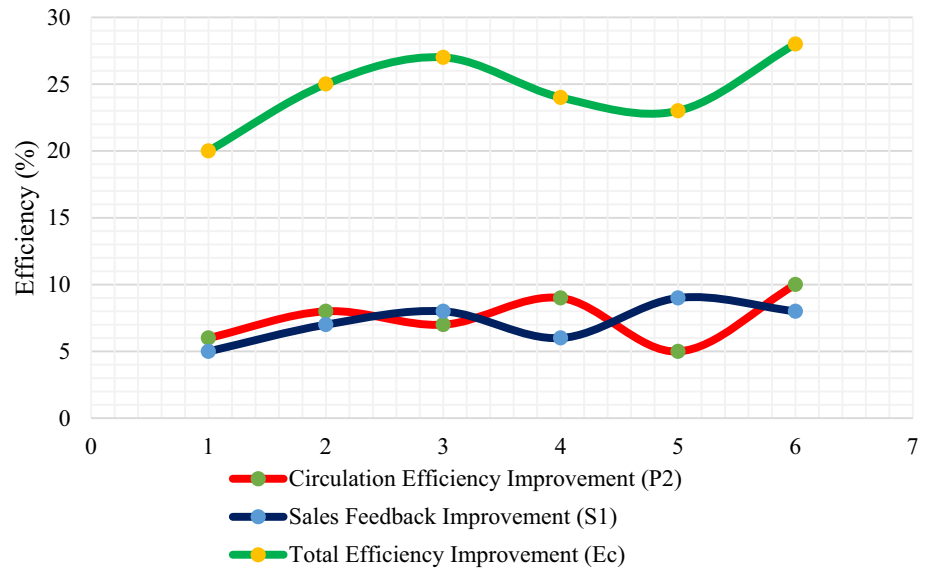


Fig. 10 Cost improvement (Ci) overview for projects

efficiency improvement (Ec) throughout significant areas. The combined effect of increases in $P1$, $P2$, and $S1$ drives overall operational excellence and performance throughout these critical zones, resulting in a cumulative total efficiency gain. Each row represents a specific region,

Table 6 Analysis of comprehensive optimization (Os) factors for different projects

Finishing improvement (Fm)	Assembly integration improvement (As)	Inventory improvement value (In)	Total comprehensive optimization (Os)
30	25	20	75
35	28	22	85
28	20	18	66
32	26	19	77
33	30	21	84
29	22	17	68

displaying their respective improvements in Production, Circulation, and Sales Feedback. For instance, the first region exhibits Production Efficiency Improvement of 18, Circulation Efficiency Improvement of 15, and Sales Feedback Improvement of 13, resulting in a Total Efficiency Improvement (Ec) of 46. Figure 11 enables a comparative analysis among regions, illustrating how variations in Production, Circulation, and Sales Feedback improvements contribute to the overall Total Efficiency Improvement. It provides insights into regional strengths and weaknesses regarding different efficiency elements, allowing assessment and comparison of their impacts on the cumulative efficiency enhancement across the key regions.

The Regional Analysis of Production Efficiency Improvement ($P1$) is categorized into Low, Middle, and High segments. Each row represents a specific region classified within these segments. For instance, in the Low Segment (0–40), the first row displays a Production Efficiency Improvement of 4, while the subsequent rows demonstrate variations in improvements within this segment. The Middle Segment (40–80) and High Segment (80–100) also exhibit their respective improvement values. It provides a clear visualization of how different regions, classified based on their initial efficiency levels, progress in Production Efficiency Improvement over consecutive instances, offering insights into improvement trends across varied segments. Figure 12 illustrates a segmented examination of production efficiency across regions, showcasing disparities and patterns of improvement among low, middle, and high-efficiency components.

An analysis of Comprehensive Optimization (Os) components divided into Low, Middle, and High segments for Finishing (Fm), Assembly (As), and Inventory (In) shows

how each component's level of technological integration, sophistication, and efficiency increases over time, impacting overall system performance and operational excellence. Each row denotes specific elements within these segments. For instance, in the Finishing category, the Low Segment displays values of 12.9, whereas the Middle and High Segments show values of 18, 20, 25, and 28, respectively. Similar patterns are observed in Assembly and Inventory categories across the segments. To enable a segmented examination of optimization elements, illustrating differences in values between Low, Middle, and High segments for Finishing, Assembly, and Inventory components. These elements vary across different efficiency segments, offering a comparative view of the improvements within each segment. Such segmentation aids in understanding the varying degrees of optimization achievements in distinct segments, providing insights into areas of strength and potential improvement across different optimization elements. Figure 13 presents a segmented examination of optimization elements, illustrating differences in values between the Low, Middle, and High segments.

The findings from the extensive analysis of efficiency, cost, comprehensive optimization, and service refinement elements within logistics operations offer profound insights into enhancing sustainability within the Sustainable Development (SD) framework. The study, based on survey data from 302 logistics enterprises, utilized sophisticated analytical tools like MATLAB21.0b and the Volar model to assess and quantify various optimization elements meticulously. Efficiency Improvement Elements (EC) analysis revealed varying improvements across segments, indicating a disparity in performance among production, circulation, and sales feedback. The segmentation enabled a nuanced understanding of the diverse levels of

advancement within logistics operations, emphasizing areas that require greater attention for enhancement.

Similarly, Cost Improvement Elements (Ci) analysis showcased distinct improvements across logistics sectors, highlighting significant gains in transportation and raw material costs in higher segments. This understanding of cost optimization at different levels elucidates critical areas for cost-saving measures and resource allocation strategies. The Comprehensive Optimization Elements (Os) analysis demonstrated escalating gains across low to ultra-maximum segments for finishing, assembly integration, and inventory elements. This tiered approach revealed the varying degrees of improvement within logistics operations, emphasizing the need for tailored optimization strategies based on the segment's performance. The Service Optimization Elements (So) analysis, represented by improvement coefficients, depicted a consistent upward trend, showcasing continuous advancements in service optimization over various projects. This illustrates the positive impact of improvement efforts on refining service quality within logistics frameworks. Collectively, these analyses underscore the importance of segmenting and understanding different facets of logistics operations for holistic improvement. The insights garnered from this comprehensive provide a roadmap for devising tailored strategies to optimize efficiency, reduce costs, enhance complete logistics elements, and refine service quality within the Sustainable Development framework. Implementing targeted interventions based on these insights can significantly contribute to sustainable and efficient logistics operations, aligning with the goals of sustainable development and environmental conservation.

Fig. 11 Efficiency improvement (Ec) breakdown for key regions

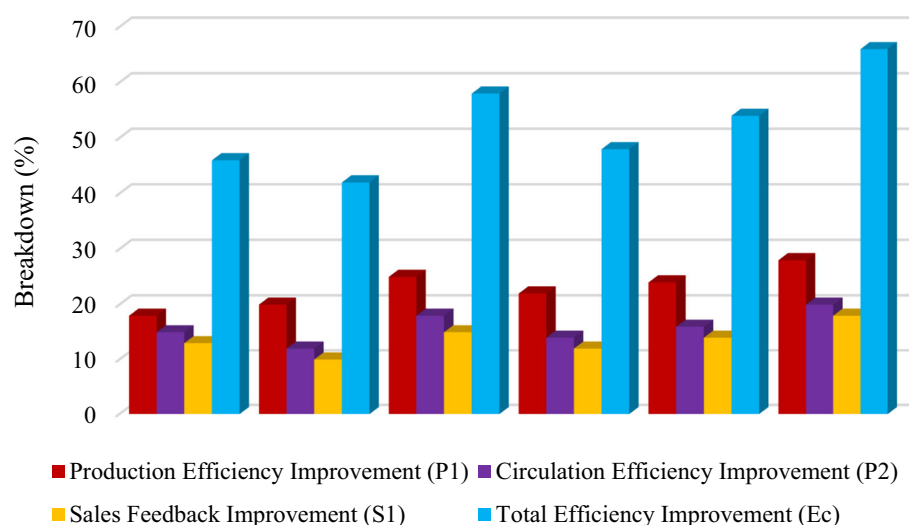


Fig. 12 Regional analysis of production efficiency improvement (P1)

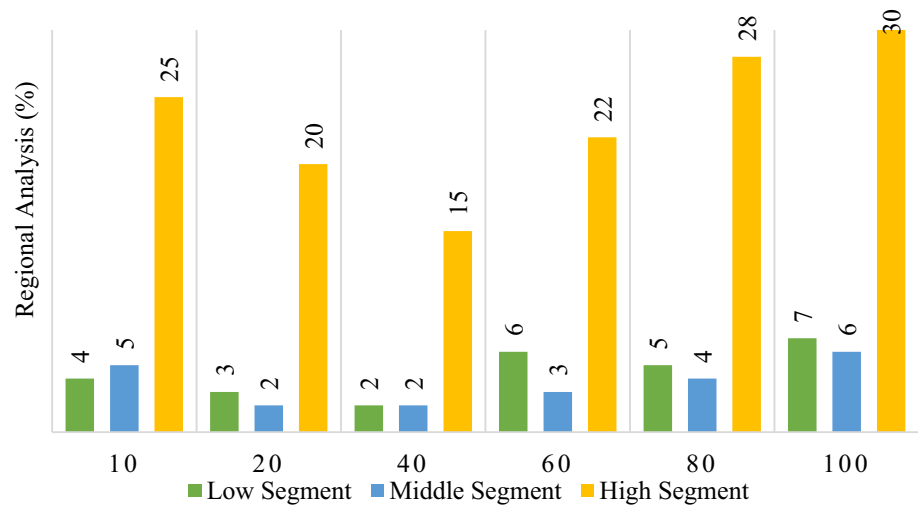
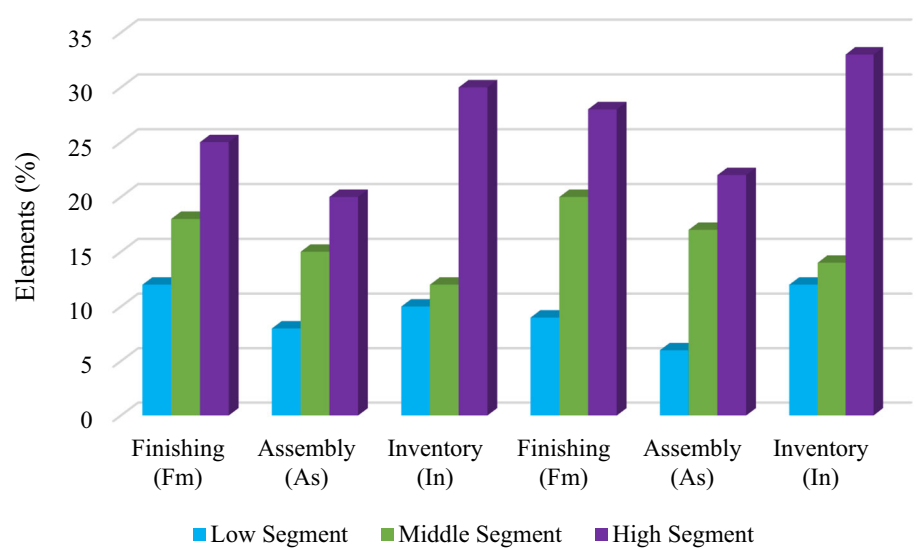


Fig. 13 Breakdown of comprehensive optimization (Os) elements



5 Conclusion

The gathered data and analyzed samples paved the way for identifying and categorizing the fresh direction for SD logistics management mode optimization. This process confirmed the mode's effectiveness by observing model results and patterns. It aimed to rectify four critical issues related to logistics management optimization: delivery efficiency, raw material costs, one-stop solutions, and service information evaluation. To tackle these, four specific analysis functions were formulated: efficiency improvement, cost improvement, comprehensive optimization, and service enhancement factors. Utilizing MATLAB 21.0b software, the performed model analyses primarily relied on the Volar model area analysis. Efficiency improvement elements within the SD logistics management module saw significant optimization through the sustainable

development of the ecological cycle. Similarly, cost improvement factors underwent effective optimization via the enhancement of the SD logistics management model's efficiency module within the sustainable development environmental cycle. Concerning comprehensive optimization elements, the efficiency module within SD logistics management exhibited effective optimization and improvement through sustainable ecological cycle development. The analysis revealed the service improvement coefficient of $S(n - 1)$ to be 0.88 ± 0.11 , while the overall service situation coefficient S_n stood at 0.89 ± 0.11 , signifying an improvement range between 88 and 100%. This suggests a noticeable and pertinent improvement effect. The analysis of the case study, focusing on optimizing SD logistics management mode, revealed promising development avenues through ecological cycle background establishment and capacity enhancements. The innovative mode

secured empirical analysis certification and gained widespread recognition from 309 logistics management enterprises in the market. These findings hold considerable empirical value and demonstrate significant correlations.

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Data and material availability Access to the data supporting the research findings can be obtained by contacting the corresponding author for further information or inquiries.

Declarations

Conflict of interest The author confirms the absence of any conflicting interests that could potentially influence the research outcomes or interpretations.

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